Deepu Vishwakarma

Varanasi, Uttar Pradesh
+91-6393323419 deepuiiituna@gmail.com LinkedIn Github

Education

Indian Institute of Information Technology Una **2023 - Present**
B.Tech in Computer Science and Engineering, CGPA: 7.84 *Una, Himachal Pradesh*

Delhi Public School **2022**
Senior Secondary - 90.2% | Secondary - 93.6% *Varanasi, Uttar Pradesh*

Experience

ApnaGharAshram | *Software Developer Intern* **May '26 – Jun '26**

- Engineered a secure, role-based admin dashboard with access control, empowering non-technical staff to manage website content and reducing dependency on developers for daily updates.
- Integrated Supabase for media storage, retrieval, and real-time synchronization, powering a gallery with uploads managed through the admin panel.
- Refactored legacy components into reusable Next.js modules, improving maintainability and page load times.

Projects

KawaiCrafts | *Node.js, Express, TypeScript, PostgreSQL (Neon), Drizzle ORM, Zod, Razorpay, Supabase* **GitHub**

- Architected a REST API in Express and TypeScript on serverless PostgreSQL (Neon) with Drizzle ORM and a shared schema covering users, products, orders, downloads, and wishlists; deployed on Vercel serverless functions.
- Implemented stateless Google OAuth 2.0 (PKCE via Arctic) with short-lived state cookies, JWT sessions in HttpOnly, Secure, SameSite cookies, and role-based admin access enforced through middleware.
- Built the payment pipeline with Razorpay/Stripe: orders created as `pending`, HMAC-SHA256 signature verification on callbacks/webhooks, then order completion and download-rights grant on success.
- Delivered purchased PDFs through Supabase Storage with access gated by ownership records; validated all request payloads with Zod and handled multipart uploads with Multer.

PayBack | *Node.js, Express, MongoDB (Mongoose), Redis, JWT, Stripe Connect, Razorpay, Cloudinary* **GitHub**

- Developed an MVC-style Express REST API with MongoDB models for users, groups, expenses, and transactions, JWT authentication, bcrypt password hashing, and Helmet/CORS/Morgan middleware for security and logging.
- Implemented a custom graph-based debt simplification algorithm that minimizes the number of settlement transactions within a group, supporting both equal and unequal bill splits.
- Built payment modules with Stripe Connect and Razorpay for real-time settlements, and file uploads to Cloudinary via Multer; integrated Redis (ioredis) for caching.

SAGE | *Node.js, Express, Socket.io, Multer, Python, YOLOv8, Gemini API* **GitHub**

- Designed an event-driven Node.js/Express backend with Socket.io for instant detection alerts, and a Multer-based pipeline for multipart thermal image uploads with persistent metadata logging.
- Integrated REST APIs with a dedicated Python YOLOv8 inference service for real-time aerial object detection, and used the Gemini API to turn raw detections into plain-English situation summaries streamed live to operators.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, C, C++
Backend: Node.js, Express.js, REST APIs, Socket.io, JWT, OAuth 2.0 (PKCE), Zod, Multer, Microservices Integration
Databases & Storage: PostgreSQL (Neon), MongoDB, Redis, Drizzle ORM, Mongoose, Supabase Storage, Cloudinary
Payments & Security: Stripe, Stripe Connect, Razorpay, HMAC Signature Verification, bcrypt, Helmet, RBAC
Developer Tools: Git, GitHub, Vercel, VS Code, Linux
AI/ML: LLM APIs (Gemini), YOLOv8
Core Subjects: Data Structures and Algorithms, DBMS, Operating Systems, Design and Analysis of Algorithms

Positions of Responsibility

PicXcel, IIIT Una **Sep '23 – Present**
Executive Member *IIIT Una, Himachal Pradesh*

- Organized 3+ workshops and events and acted as the core coordinator for PicXcel, the photography club of IIIT Una.

AAVESH, IIIT Una **Sep '23 – Present**
Core Advisor *IIIT Una, Himachal Pradesh*

- Organized 5+ workshops and events, served as the core coordinator for AAVESH (Technical Club), and led the development of a model rocket for TechTatva.

Achievements

- Secured a Bronze Medal at Innovation Expo, MERAKI'25, IIIT Una – a technology exhibition and innovation competition featuring student-led innovations.
- Designed and demonstrated a drone-based project (SAGE) at Sky Circuit, MERAKI'26, IIIT Una – a flagship drone-building and demonstration event.
- Knight at LeetCode: solved 700+ problems (Rating: 1865).
- 2 Star rating (1418) on CodeChef.